

Bill of Materials - VBF-T5R1-BOM-01

VBF-T5R1 delivery package - generated 2026-08-25

Bill of Materials - VBF-T5R1-BOM-01 - sheet 'BOM'

#	Component	Material	g per cell	kg per kWh	Formula	Source
1	Positive electrode - active material	NMC811 composite (96 wt% split)	16.168	0.7271	16.842×0.96	electrode layer mass mechanical (calc-energy layer_kg_m2 x area); 96:2:2 mass split = literature default (estimate - parameter set models single-phase composite)
2	Positive electrode - conductive additive	carbon (2 wt% split)	0.337	0.0151	16.842×0.02	same annotation as #1
3	Positive electrode - binder	PVDF-class (2 wt% split)	0.337	0.0151	16.842×0.02	same annotation as #1
4	Negative electrode - active material	graphite(-SiOx) composite (96 wt% split)	9.506	0.4275	9.902×0.96	same annotation as #1
5	Negative electrode - conductive additive	carbon (2 wt% split)	0.198	0.0089	9.902×0.02	same annotation as #1
6	Negative electrode - binder	SBR/CMC-class (2 wt% split)	0.198	0.0089	9.902×0.02	same annotation as #1
7	Separator	polyolefin-class, 8 um / porosity 0.55 / 397 kg/m3	0.147	0.0066	$8e-6 \times (1-0.55) \times 397 \times 0.1027 \times 1000$	mechanical from bridge/r7b_negR261.json + parameter-set density
8	Electrolyte	EC-based + 1 M LiPF6, 9.53 g (NOT in contract cell mass)	9.532	0.4287	pore vol $7.943 \text{ cm}^3 \times 1.2 \text{ g/cm}^3$	pore volume mechanical from layer geometry; density 1.2 g/cm^3 = literature default (parameter set lacks the key)
9	Positive current collector	Al foil 8 um / 2700 kg/m3	2.218	0.0997	$8e-6 \times 2700 \times 0.1027 \times 1000$	mechanical (bridge params)
10	Negative current collector	Cu foil 6 um / 8960 kg/m3	5.521	0.2483	$6e-6 \times 8960 \times 0.1027 \times 1000$	mechanical (bridge params)
11	Enclosure / tabs	Not modeled	None	None	None	honest annotation: outside pure-simulation boundary
None	TOTAL (contract caliber, electrolyte excluded)	None	34.631	1.5574	sum of mechanical rows	calc-energy mass_kg = 0.0346308 kg (matches)
None	TOTAL incl. electrolyte (reference)	None	44.163	1.9861	contract + electrolyte	reference caliber only
None	Total energy (Chen2020 / OKane2022)	None	22.2364 / 22.2024 Wh	None	1C time-integrated V*I	cell/r7b_{chen,okane}_energy.json
None	Energy density (Chen2020 / OKane2022)	None	642.1 / 641.1 Wh/kg	None	E / contract mass	cell/r7b_{chen,okane}_energy.json